Agentic Legal Case Research

Yinuo Zhao / MIT 1.125 / Individual project proposal

A research workspace for lawyers

A lawyer supplies facts and a legal question. An agent discovers and compares cases, then uses optional lawyer feedback to guide further research.

Research input

Fact pattern and legal issue
Target court and search jurisdictions
Optional starting case or citation

Agent contribution

Search, read, and refine queries
Inspect supporting and contrary cases
Explain relevance with source passages

Lawyer output

Ranked, inspectable case shortlist
Saved cases, notes, tags, and excerpts
Comparison and CSV export

Research question

Does iterative search with lawyer feedback find more useful cases than fixed search at a comparable cost?

Problem and audience

Lawyers need cases that address the right issue, facts, court, and stage of litigation.

Target user and task

A lawyer investigating a specific matter, preparing an argument, or checking whether an existing case is useful.

Example research request

“Find dismissal cases following an employee’s safety complaint. Focus on whether the decision-maker knew about the complaint.”

Illustrative fact pattern, not legal advice

Discovery

Different terminology can hide useful cases.
Search reformulation and citation trails create leads.

Applicability

Similar facts may involve a different issue or ruling.
Show the issue, reasoning, court, and procedural stage.

Counterarguments

Unfavorable outcomes can be highly relevant.
Seek distinctions and cases that challenge the argument.

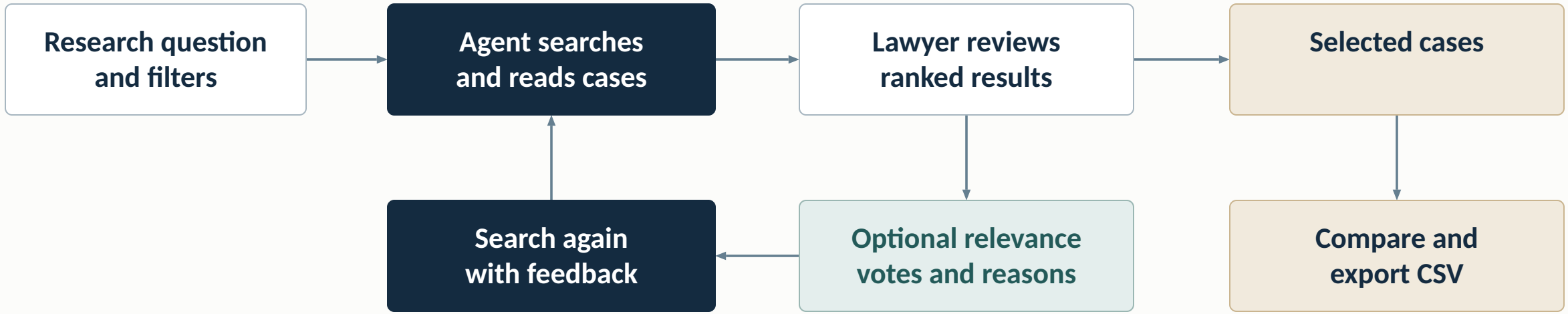
Verification

Lawyers need to check the underlying opinion.
Link each comparison to the actual supporting text.

Design hypothesis: validate the workflow and relevance criteria with a lawyer or law-student reviewer.

Proposed solution

Lawyer-guided search with a separate, persistent collection of selected cases.



Repeat as often as desired

Notes, tags, excerpts persist across rounds

Search controls

Jurisdiction, court, dates, case citation, keywords, and exact phrases.
Advanced filters depend on metadata.

Evidence in every result

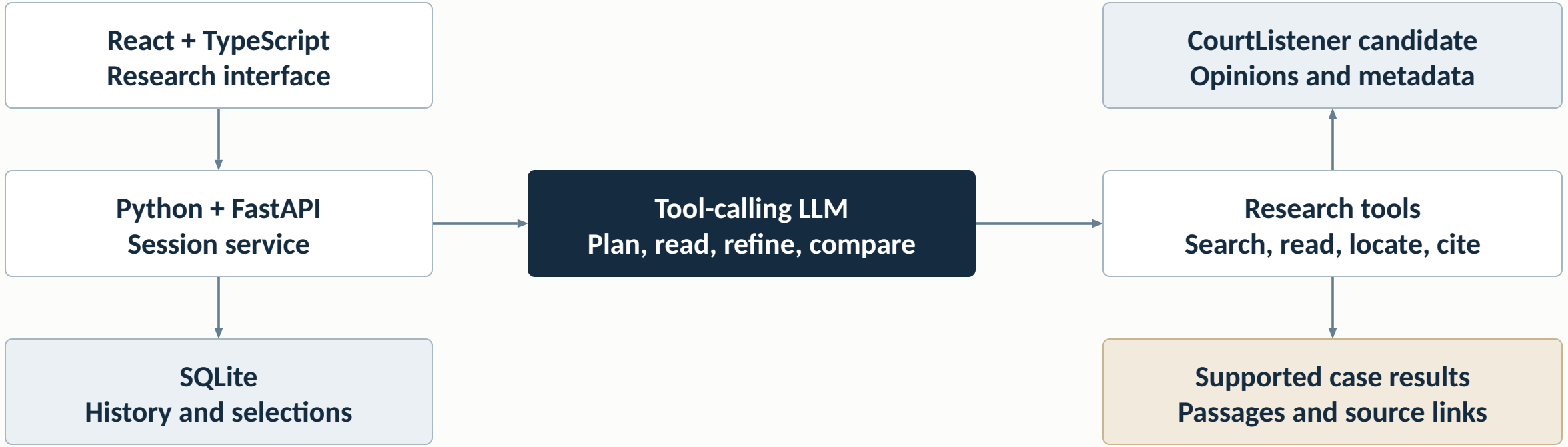
Facts, issue, holding, and reasoning.
Explain material differences.
Open excerpts with source locators.

Feedback semantics

Relevant can include unfavorable.
Saving and voting stay separate.
The latest judgment takes precedence.

Approach

One tool-using agent, structured session state, and a bounded case collection.



Retrieval and data

Check access and full-text coverage.
Use stable IDs and permitted caching.
One U.S. topic and court subset.

Agent control

Read sources and pursue citation leads.
Budget each research round.
Use feedback to refine searches.

Libraries and model selection

HTTPX, Pydantic, pytest candidates.
Pilot a hosted LLM before selection.
Technology choices are provisional.

Scope and risks

An individual course prototype with explicit coverage limits and source-based review.

Committed scope

One topic and bounded case collection
Filters and known-case lookup
Results linked to source passages
Unlimited user-initiated rounds
Saved cases, comparison, CSV export

Next, if data and time permit

Bidirectional citation exploration
Advanced filters and passage feedback
A brief from selected cases

Deferred

Comprehensive treatment verification, automatic authority labels, accounts, and broad coverage

Data access and coverage

Validate full-text access, terms, and limits first.
Reduce the corpus if metadata or access is incomplete.

Legal interpretation and hallucination

Require supporting passages and expose uncertainty.
Distinguish a procedural ruling from a final outcome.

Limited legal expertise

Seek lawyer or law-student judgments and a walkthrough.
Without expert review, report findings as exploratory.

Cost and repetitive search

Cap work within each round and track latency/cost.
Deduplicate results and disclose diminishing returns.

Treatment status: “not checked” unless a suitable source verifies it.

Plan and success criteria

A working end-to-end demo plus a controlled comparison on approximately 10–15 research tasks.

Milestones and deliverables

1 Confirm data feasibility

Check access, coverage, and reuse terms.
Define the domain and benchmark questions.

2 Build retrieval and workspace

Implement filters, source inspection, saving, notes, excerpts, and CSV export.

3 Add agent and feedback loop

Refine queries, pursue leads, and preserve state.
Support revised feedback and repeated rounds.

4 Evaluate and demonstrate

Run comparisons, inspect errors, and deliver setup instructions, example sessions, and report.

Three comparison conditions

A Fixed search and retrieval
B Iterative agent search without feedback
C Iterative agent search with relevance feedback

Same corpus and comparable retrieval budgets.
Judge later results separately from feedback examples.

What counts as success

Functionality: full workflow, persistent state, correct export.
Evidence: quotes and locators match the source opinion.
Retrieval: top-5 precision, new relevant and contrary cases.
Efficiency: latency, tool calls, and estimated model cost.

Improvement over the baseline is a hypothesis to test.
Report negative findings and expert-review limitations.